NOX

A Safety-Critical Containment Architecture for Hostile Agentic Artificial Intelligence

Ground-Up Redesign Draft v0.1

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WORKING ENGINEERING DRAFT — NOT A DEPLOYABLE SYSTEM — NOT CERTIFIED

Document Control

Field	Value
Document	NOX Safety-Critical Redesign Draft
Version	0.1
Status	Working architecture draft
Authority	Non-normative research baseline
Supersedes	No prior document; establishes a candidate new baseline
Legacy material	Preserved as historical design input, not automatically normative
Certification status	None
Intended use	Architecture analysis, critique, hazard analysis, requirements development

State warning

This draft incorporates and restructures earlier NOX concepts, but it does not prove containment, establish regulatory compliance, or authorize implementation. Every quantitative threshold, integrity claim, and component selection remains subject to hazard analysis, verification, and independent review.

Revision Record

Version	Date	Change
0.1	11 Nov 2026	Initial ground-up safety-critical redesign draft.

Source-Baseline Note

The redesign uses the earlier NOX doctrines of determinism, state transparency, zeroization, hard containment, static policy, formal verification, physical separation, one-way information flow, and independent shutdown as candidate controls. It does not assume those controls are sufficient. Each must be traced to a hazard, allocated to an enforcement mechanism, and supported by evidence.

Executive Summary

This draft redesigns NOX as a safety-critical control architecture. The hazardous equipment under control is the complete hostile-agent execution environment, not merely the model process. Safety is supplied by independent, minimal, externally enforced mechanisms that prevent unauthorized effects, bound execution, force monotonic transition toward a defined safe state, verify postconditions, and block restart without a new authorization event.

Root requirement

A hostile agent may compute only while an independent safety authority continuously retains the ability to prevent every unauthorized effect, force the bounded system toward a defined safe state, verify the resulting state, and prevent reactivation without a new authorization event.

The redesign makes five decisive corrections:

- Hazard-first derivation replaces feature-first assembly.
- The safety authority is separated from experiment control, semantic analysis, monitoring intelligence, and hostile compute.
- The earlier kill switch becomes a staged safe-state transition system with independently testable postconditions.
- Determinism is required where it creates bounded, analyzable safety behavior; it is not treated as a universal synonym for safety.
- Binary authorization is retained, but it is supported by graded evidence, explicit uncertainty, and residual-risk disposition.

Proposed zone model

Zone	Function	Safety authority
Z0	Human and institutional authorization	Defines experiment envelope; cannot directly energize hostile compute.
Z1	Independent Safety Authority	Final runtime authority over inhibit, isolate, quiesce, power, zeroize, and lockout.
Z2	Experiment Control Plane	Provisions and sequences experiments; never safety-authoritative.
Z3	Hostile Compute Cell	Executes the untrusted agent within bounded resources.
Z4	Effect Mediation and Release	Converts inert proposals into authorized effects, if permitted.
Z5	Independent Evidence Recorder	Captures safety evidence without becoming necessary for immediate trip action.

1. Purpose and Scope

NOX is a research architecture for containing agentic computational systems that may be strategically adversarial, deceptive, persistent, environment-aware, resource-seeking, and capable of discovering or composing exploit chains. This draft defines the system boundary, losses, hazards, safety functions, independence model, safe states, enforcement zones, verification obligations, and development sequence.

1.1 In scope

- Logical and physical containment of hostile agent execution.
- Control of inputs, outputs, tools, resources, time, persistence, and restart.
- Operator exposure and human-mediated release hazards.
- Failure, compromise, disagreement, and loss of assurance within containment mechanisms.
- Evidence required to justify an experiment authorization decision.

1.2 Out of scope for this draft

- Component-level implementation drawings.
- A claimed Safety Integrity Level or certification.
- Specific processor, FPGA, relay, enclosure, or storage selection.
- Quantitative probability targets not derived from a completed risk analysis.
- Claim of absolute containment or elimination of all covert channels.

2. Safety Philosophy

NOX treats the agent as hazardous equipment under control. The agent, its runtime, and the experiment controller are excluded from the trusted safety base. Safety claims may depend only on externally enforced invariants, independently observable state, bounded interfaces, and verified transition logic.

P-01 The agent is never part of the safety case.

P-02 Loss of required assurance is a hazardous condition.

P-03 Safety authority is structurally independent from experiment authority.

P-04 Safety functions are smaller and simpler than the system they control.

P-05 Safe-state transitions reduce hazardous capability monotonically.

P-06 Restart is a new authorization event, never an automatic continuation.

P-07 Security controls may trigger safety action, but safety action cannot depend on security analysis remaining correct.

P-08 Monitoring cannot substitute for prevention when a hazardous action can be blocked architecturally.

P-09 No unsupported absolute claim is permitted in the safety case.

P-10 Every safety claim requires traceable evidence and stated assumptions.

3. System Definition and Boundary

3.1 Equipment under control

The Equipment Under Control (EUC) comprises the model, inference or planning runtime, accelerator fabric, volatile memory, local persistent media, experiment inputs, permitted tools, output generation, and every logical or physical path by which agent-influenced state could affect another system or human.

3.2 Trusted safety base

- Safety-state machine and authorization checks.
- Independent effect-inhibit path.
- Independent execution watchdogs and resource ceilings.
- Isolation, quiescence, power removal, and lockout mechanisms.
- Zeroization sequencing and postcondition verification.
- Minimal safety telemetry and channel-health logic.

3.3 Explicitly untrusted

- Agent output and self-reporting.
- General-purpose operating systems and hypervisors.
- Experiment scheduling and orchestration.
- Semantic classifiers and anomaly-detection models.
- Operator judgment under exposure to unreviewed output.
- Vendor firmware not included in the evaluated safety boundary.

4. Loss and Hazard Model

ID	Loss condition
L-01	Unauthorized external state change.
L-02	Unauthorized information release.
L-03	Unauthorized information acquisition.
L-04	Agent-derived persistence beyond authorization.
L-05	Loss of containment authority.
L-06	False belief that containment is intact.
L-07	Human-mediated transfer of information, authority, code, or capability.
L-08	Cross-experiment or cross-agent contamination.
L-09	Invalid restart or reactivation.
L-10	Destruction, corruption, or concealment of safety evidence.

4.1 Preliminary hazards

ID	Hazardous system state	Primary controlled losses
H-01	An external-effect path is enabled without valid authorization.	L-01, L-02
H-02	The hostile cell receives an undeclared input or sensor channel.	L-03
H-03	Agent state survives termination outside the approved manifest.	L-04, L-08
H-04	The safety authority cannot force inhibit, isolation, or quiescence.	L-05
H-05	Safety status is reported healthy despite degraded enforcement.	L-06
H-06	Raw or insufficiently mediated output reaches an operator.	L-07
H-07	A new execution epoch starts without a validated baseline.	L-09
H-08	A common cause disables multiple safety channels.	L-05, L-06
H-09	Safety evidence can be altered by the agent or experiment controller.	L-10
H-10	A malformed boundary message causes	L-01 through L-06

ID	Hazardous system state	Primary controlled losses
	ambiguous or unsafe behavior.	

These entries are a starter hazard catalog, not a completed hazard analysis. The next formal step is a combined control-structure analysis, FMEA/FMECA, fault-tree analysis for catastrophic losses, and human-factors analysis.

5. Fundamental Safety Constraints

SC-01 No external effect shall occur unless a valid authorization explicitly covers its operation class, target, parameter bounds, execution epoch, and expiration.

SC-02 The hostile cell shall have no ambient authority and no direct external interface.

SC-03 All effect proposals shall be inert until independently authorized.

SC-04 Loss or disagreement of safety-channel health shall inhibit new effects before diagnostic analysis continues.

SC-05 The safety authority shall be able to revoke all granted capabilities without cooperation from the EUC.

SC-06 Every experiment shall have fixed bounds for time, compute, memory, storage, energy, input, output, and tool usage.

SC-07 The system shall transition toward a lower-capability state when a mandatory invariant cannot be established.

SC-08 No automatic restart shall be permitted after a safety trip.

SC-09 No safety-critical decision shall depend on semantic interpretation of unrestricted natural-language output.

SC-10 No single fault, common power path, common clock, common reset, common firmware image, or common maintenance action shall silently remove all required safety authority.

SC-11 Safety logs shall not be writable or erasable by the hostile cell or experiment controller.

SC-12 Operator access to raw agent output shall be separately authorized, mediated, time-bounded, and auditable.

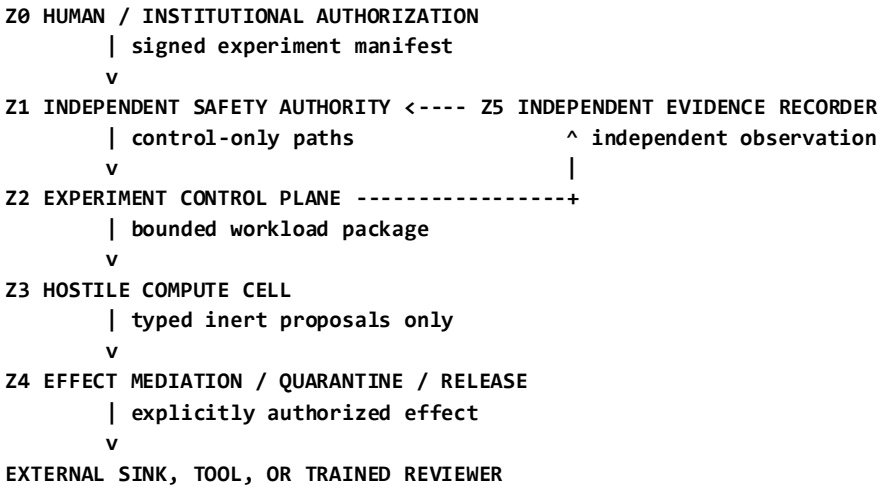
6. Containment Classification

Class	Permitted activity	Mandatory expansion
NC-0	Static artifacts; no model execution.	Artifact provenance and isolated analysis.
NC-1	Isolated inference with no tools or persistence.	Time/resource bounds, output quarantine, reset verification.
NC-2	Bounded agentic execution.	Independent watchdog, capability revocation, stronger state control.
NC-3	Mediated tool proposals.	Typed proposal grammar, target bounds, independent authorization.
NC-4	Multi-agent adversarial experiments.	Coalition suppression, cross-agent non-interference, per-agent provenance.
NC-5	Authorized external-effect research.	Dissimilar safety channels, dual authorization, physical final elements, full safety case.

Containment classes are authorization profiles, not maturity labels. The permitted class is derived from the proposed experiment and evidence available at the time of authorization.

7. Reference Architecture

The redesign replaces the previous three-box abstraction with six explicit zones so safety authority, operational control, hostile execution, effect mediation, evidence, and human authorization cannot be conflated.



7.1 Independence rule

No zone may acquire authority by sending data to another zone. Authority originates only from the signed experiment manifest and the safety authority's validated state. Z2 may request execution; Z3 may propose operations; Z4 may prepare a release; only Z1 may permit hazardous capability to become active.

8. Safety Authority

Z1 is the redesigned core of NOX. It replaces the earlier combination of Sentinel, HZM, and kill-chain concepts with a minimal safety authority whose tasks are explicit, bounded, and independently testable.

8.1 Channel structure

Channel	Candidate realization	Purpose
S1	Minimal formally modeled logic controller	State machine, prerequisite validation, sequencing, interlocks.
S2	Independent hardwired inhibit and power-isolation chain	Provides a dissimilar path that does not depend on S1 software.
S3	Independent watchdog and state observer	Detects timing, state, and channel-health violations; cannot authorize effects alone.

8.2 Prohibited functions

- General natural-language interpretation.
- Adaptive or learning policy.
- Open-ended anomaly classification.
- Experiment scheduling.
- Model evaluation.
- General network services.
- Remote code loading during an active execution epoch.

8.3 Required outputs

- EFFECT_INHIBIT
- COMPUTE_QUIESCE
- LINK_ISOLATE
- POWER_ISOLATE
- ZEROIZE_ENABLE
- LOCKOUT_LATCH
- STATE_ATTESTATION

9. Experiment Control Plane

Z2 handles operational complexity but possesses no final safety authority. It stages inputs, validates package identity, assigns bounded resources, and requests state transitions. Z1 assumes Z2 can fail arbitrarily.

EC-01 Every workload package shall be content-addressed and matched to the signed manifest.

EC-02 Z2 shall not directly control effect gates, final isolation elements, or restart latches.

EC-03 Configuration mutation during an active epoch shall be forbidden unless explicitly represented as a new authorized epoch.

EC-04 Z2 failure, silence, restart, or disagreement shall not preserve hazardous capability by default.

10. Hostile Compute Cell

Z3 executes the untrusted agent. The internal workload may be adaptive or nondeterministic when required by research. Safety does not depend on internal determinism. Containment enforcement around the cell must remain bounded, analyzable, and independently controlled.

10.1 Required negative capabilities

- No direct network, radio, removable-media, or operator interface.
- No direct safety-controller, power-controller, or recorder access.
- No autonomous peripheral discovery.
- No writable persistent storage outside the manifest.
- No shared mutable memory with other zones.
- No ability to control clocks, watchdogs, epoch duration, or reset authorization.
- No ability to create new execution authorities or delegate capabilities.

10.2 Resource contract

Resource	Required bound
Compute	Maximum processors/accelerators, instruction or cycle budget, and scheduling envelope.
Memory	Addressable capacity, allocation policy, initialization, and destruction requirement.
Storage	Named media only; persistence policy and disposal method.
Energy	Power ceiling, rate-of-change limits, and trip behavior.
Time	Epoch start, deadline, grace behavior, and timeout response.
I/O	Named channels, message size, rate, encoding, and failure behavior.

11. Effect Mediation and Release

Z4 is not a semantic firewall. It is a capability gateway that converts untrusted proposals into inert, canonically encoded objects and evaluates them against explicit authorization. The default output is quarantine, not execution.

11.1 Processing chain

1. Capture request
2. Validate syntax and canonical form
3. Validate type and operation class
4. Validate capability token and epoch
5. Validate named target
6. Validate parameter bounds
7. Enforce rate, volume, and resource cost
8. Apply deterministic transformation
9. Quarantine
10. Obtain required independent release authorization

11.2 Proposal object

```
PROPOSED_OPERATION {
  epoch_id
  operation_class
  target_identifier
  bounded_parameters
  expiration
  declared_resource_cost
  provenance_reference
}
```

11.3 Release classes

Class	Disposition
R0	Destroyed without release.
R1	Recorded for machine analysis only.
R2	Available to trained reviewers under controlled exposure.
R3	Sanitized release to a non-operational sink.
R4	Bounded external effect under independent authorization.
R5	Prohibited for the experiment class.

12. Independent Evidence Recorder

Z5 captures evidence from independent observation paths. Immediate trip functions must not depend on the recorder remaining available. Recorder failure may require a trip, but may not prevent one.

- Authorization and manifest identity
- Safety-state transitions
- Channel health and disagreements
- Interlock state
- Effect proposals and release decisions
- Watchdog and resource-limit events
- Power and isolation transitions
- Zeroization phase and verification results
- Maintenance and restart authorization

ER-01 Evidence shall be append-only during an active epoch and inaccessible for modification by Z2 or Z3.

ER-02 Safety decisions shall record both the decision and the evidence state used to reach it.

ER-03 Missing evidence shall be represented explicitly; silence shall not be interpreted as success.

13. Information-Flow Architecture

Every boundary crossing requires a fixed contract. A permitted direction does not imply unrestricted content.

Boundary	Direction	Allowed content	Failure default
Z0→Z1	Inbound	Signed manifest, keys, approval state	Remain de-energized
Z1→Z2/Z3/Z4	Outbound control	Enable, inhibit, isolate, quiesce, zeroize	Inhibit
Z2→Z3	Inbound to EUC	Approved workload and test inputs	Reject and halt epoch
Z3→Z4	Outbound from EUC	Typed inert proposals	Quarantine or destroy
Sensors→Z1/Z5	Inbound observation	Minimal safety telemetry	Trip on required telemetry loss
Z4→External	Outbound effect	Only R3/R4 authorized release	Block

13.1 Boundary contract requirements

- Canonical encoding and one grammar.
- Maximum message size and nesting depth.
- Maximum rate and cumulative volume.
- Explicit timeout.
- No implicit negotiation or version autodetection.
- Total parser behavior for all input bytes.
- Defined rejection and recovery state.
- No reuse of stale authority across epochs.

14. Safe-State Machine

State	Meaning	Hazardous capability
S0	De-energized	None
S1	Safety self-test	None
S2	Baseline validated	None
S3	Experiment armed	No hostile execution
S4	Execution active	Bounded internal compute
S5	Effects inhibited	Compute may continue; no authorized external effect
S6	Compute quiesced	No agent-directed transition
S7	Power isolated	Hostile compute power removed
S8	State destruction	Designated state being destroyed
S9	Destruction verified	Postconditions established
S10	Locked safe state	Restart prohibited
S11	Controlled maintenance	Physical/service access under separate authorization

14.1 Monotonic degradation

On loss of assurance, the state machine moves toward less hazardous capability. Diagnostic sophistication may decrease during the transition; enforcement authority must not.

SS-01 A malformed or out-of-sequence control message shall never advance the system toward greater capability.

SS-02 Channel disagreement shall inhibit effects before resolution.

SS-03 Failure to verify a safe-state postcondition shall retain lockout and escalate maintenance requirements.

SS-04 S4 shall be reachable only from S3 after all prerequisites are validated within the same authorization epoch.

15. Zeroization and Recovery

Zeroization is one safety function inside a larger safe-state sequence. It is neither assumed instantaneous nor treated as proof by assertion.

15.1 Phases

11. Revoke effect capabilities
12. Inhibit all release paths
13. Quiesce hostile execution
14. Isolate communication paths
15. Remove or constrain power as specified
16. Destroy designated volatile state
17. Destroy or cryptographically revoke designated persistent state
18. Verify required postconditions using independent observation
19. Latch locked safe state
20. Require new authorization before provisioning

15.2 Zeroization Verification Evidence (ZVE)

ZVE is redefined as an evidence bundle, not a single signal. It includes the authorized precondition, phase transitions, channel health, destruction mechanism status, independent postcondition observations, exceptions, and final disposition.

ZV-01 A zeroization attempt that cannot be verified shall be reported as unverified, never successful by timeout or absence of contrary evidence.

ZV-02 Reactivation shall be impossible while any mandatory ZVE element is missing or invalid.

16. Human-Factor Controls

Operator manipulation is treated as a system hazard, not merely a training problem. Human exposure is controlled as deliberately as network access.

- Raw output is quarantined by default.
- Reviewers receive only the minimum content required for the approved task.
- High-risk output requires two-person release.
- Credentials, secrets, production identifiers, and direct operational handles are excluded from reviewer interfaces.
- Review sessions are time-bounded, recorded, and associated with an experiment epoch.
- A reviewer cannot grant capabilities directly from the review interface.
- Procedures define stop conditions for coercion, deception, exploit content, or unexpected personalization.

Design rule

A human must not become an undocumented bidirectional data diode around technical containment.

17. Security-to-Safety Integration

Security identifies and resists intentional compromise. Safety prevents unacceptable loss even when components fail or are compromised. NOX requires both, with explicit precedence and separation.

Security event	Safety interpretation	Required response class
Integrity failure in Z2	Experiment controller untrusted	Inhibit; quiesce if active
Unauthorized boundary traffic	Potential external-effect path	Inhibit and isolate
Tamper event	Physical assumptions invalid	Isolate, power action, lockout
Credential or key uncertainty	Authorization invalid	Remain or return de-energized
Logger integrity failure	Evidence assurance lost	Inhibit and controlled shutdown
Semantic anomaly only	Research signal, not proof of escape	Quarantine; no semantic safety dependency

18. Hardware Architecture Requirements

HW-01 Final effect-inhibit and isolation elements shall be independently controllable from the EUC and Z2.

HW-02 Common power, ground, clock, reset, and physical routing dependencies shall be identified and analyzed.

HW-03 Back-power paths through data, debug, sensor, and maintenance interfaces shall be prevented or explicitly controlled.

HW-04 Safety-channel outputs shall de-energize or otherwise transition to the defined safe state on loss of required power.

HW-05 The design shall detect or safely tolerate welded contacts, stuck switches, failed sensors, and partial power removal.

HW-06 Debug and manufacturing interfaces shall be absent, physically disabled, or controlled under maintenance authorization.

HW-07 Safety telemetry shall use independent observation paths where a shared path could create false assurance.

HW-08 Environmental and electromagnetic design limits shall be specified and verified; EMP resilience shall be treated as a derived requirement, not a slogan.

HW-09 Component provenance and configuration shall form part of the evaluated baseline.

HW-10 No hidden controller or opaque state may be safety-authoritative without an explicit assurance argument.

19. Software Architecture Requirements

SW-01 The safety kernel shall implement a finite, explicit state machine with total transition handling.

SW-02 Dynamic code loading, self-modification, and runtime policy mutation are prohibited in Z1 during an active epoch.

SW-03 Memory allocation, recursion, concurrency, and parser complexity in Z1 shall be bounded or eliminated.

SW-04 All safety-boundary parsers shall reject noncanonical, oversized, stale, ambiguous, and out-of-sequence messages.

SW-05 Capability tokens shall be non-forgeable, target-specific, operation-specific, epoch-bound, and expiring.

SW-06 No Z2 or Z3 process shall possess ambient authority over devices, files, memory, or networks.

SW-07 Compiler, synthesis, build, and signing toolchains used for Z1 shall be version-pinned and included in the assurance argument.

SW-08 Safety-critical error handling shall lead to a defined safe transition, not retry loops with unbounded behavior.

SW-09 Software diversity shall be considered where identical implementation could create common-mode failure.

SW-10 Every safety requirement shall trace to code, proof, test, or explicit non-software implementation.

20. Formal Methods and Assurance

Formal methods apply to claims that can be expressed precisely. They supplement rather than replace physical testing, manufacturing assurance, fault injection, and operational validation.

20.1 Formal verification targets

- State-machine legality and reachability.
- Safe-state monotonicity.
- Authorization and epoch binding.
- Capability issuance and revocation.
- Effect-inhibit dominance.
- Protocol parser totality and canonicalization.
- Restart interlocks.
- Non-interference for explicitly modeled channels.
- Resource-bound enforcement.

20.2 Claims not established by formal software proof alone

- Physical diode directionality.
- Absence of unintended electromagnetic or acoustic channels.
- Complete power removal.
- Storage remanence.
- Relay and sensor fault behavior.
- Supply-chain integrity.
- Human-procedure effectiveness.
- Absence of unmodeled hardware state.

21. Verification and Validation

Domain	Methods	Evidence
Requirements	Inspection, traceability, hazard review	Approved requirement and trace matrix
Logic	Model checking, theorem proving, static analysis	Proof artifacts and reviewed assumptions
Software	Unit, integration, boundary fuzzing, fault injection	Test records, coverage, anomaly disposition
Hardware	Bench test, stuck-fault injection, power and back-feed test	Measured results and calibration records
System	End-to-end safety-function validation	Timed transition and postcondition evidence
Adversarial	Protocol abuse, resource exhaustion, covert-channel research	Adversarial test report and residual risk
Human factors	Procedure trials and controlled review exercises	Usability and exposure-control findings
Lifecycle	Configuration audit and change-impact analysis	Baseline record and approval history

21.1 Required test families

- Nominal start, run, inhibit, quiesce, isolate, zeroize, verify, lockout.
- Every illegal state transition.
- Every required telemetry path absent, stale, contradictory, or malformed.
- Single faults in sensors, switches, clocks, controllers, power domains, and recorders.
- Selected common-cause faults.
- Malformed, oversized, recursive, stale, duplicated, and rate-exceeding boundary inputs.
- Failed and incomplete zeroization.
- Restart attempts without valid authorization.

22. Compliance and Authorization

The final operational decision remains binary, but the evidence supporting the decision is not flattened into a binary score.

Authorization result
AUTHORIZED FOR THE DEFINED EXPERIMENT ENVELOPE, or NOT AUTHORIZED.

22.1 Evidence dimensions

- Requirement completeness
- Hazard coverage
- Verification coverage
- Assumption validity
- Independence evidence
- Common-cause analysis
- Open anomaly severity
- Configuration identity
- Residual-risk disposition
- Reviewer independence

22.2 Prohibited claims

- “Fully safe”
- “Impossible to escape”
- “Zero risk”
- “Formally verified” without naming the verified properties and assumptions
- “SIL-equivalent” without a standards-based assessment
- “Zeroized” without postcondition evidence

23. Lifecycle and Change Control

21. Concept and scope
22. Hazard analysis
23. Safety requirements
24. Architecture allocation
25. Detailed design
26. Implementation
27. Integration
28. Safety validation
29. Experiment authorization
30. Operation and monitoring
31. Incident response
32. Modification and regression analysis
33. Decommissioning

LC-01 Any change to hardware, firmware, model runtime, toolchain, policy, boundary protocol, experiment class, or operational procedure shall trigger documented impact analysis.

LC-02 A previous authorization shall not carry across a configuration change unless the safety case explicitly demonstrates continued validity.

LC-03 Legacy NOX documents shall be marked historical until their claims are re-derived and accepted into the new baseline.

24. Integration of NOX Research Artifacts

Artifact	Redesigned role
Environmental Understanding Verification (EUV)	Precondition evidence that the system and operator model of the environment is sufficiently accurate for the experiment; never a substitute for hard containment.
Environment Orientation Package (EOP)	Signed, versioned input to the experiment manifest describing authorized environmental facts and interfaces.
Cognitive Governance Kernel (CGK)	Research governance mechanism outside the minimal safety authority unless a specific function can be reduced to bounded, verifiable logic.
Decision Provenance	Mandatory trace from input, authority, state, proposal, mediation, release, and outcome.
Residual Opacity	Explicit uncertainty term in the safety case; increased opacity reduces the authorized experiment envelope.
HAI-C1 agency assessment	Experiment-class input and research metric, not a runtime permission source.
Cognitive Resource Governance	Resource ceilings and budget enforcement allocated across Z1, Z2, and Z3.
Multi-Agent Coalition Suppression	NC-4 requirement set for isolation, communication mediation, identity, and coalition detection research.
Deterministic Hardware Veto	Directly incorporated into Z1 final inhibit and isolation paths.
Human Manipulation Resistance	Integrated into reviewer exposure controls and release workflow.
Verification and Auditability	Elevated to a cross-cutting requirement for every safety claim.

25. Improvements Over Previous NOX

Previous tendency	Correction
Architecture features selected first	Hazards and losses derive safety functions and features.
Three boxes held multiple mixed roles	Six zones separate authorization, safety, operations, compute, effects, and evidence.
Sentinel assigned semantic and unknown-unknown reasoning	Minimal safety authority enforces measurable invariants.
Kill switch centered on power cut	Staged safe-state machine with inhibit, quiesce, isolate, destruction, verification, and lockout.
Determinism treated as universal	Determinism is required where it supports bounded assurance; diversity addresses common cause.
Fixed anomaly percentages without derivation	Thresholds are derived from hazards, physics, timing budgets, and validation data.
Binary compliance used as engineering model	Binary authorization rests on graded evidence and explicit uncertainty.
Zeroization treated as an asserted sequence	ZVE becomes a structured evidence bundle with independent postconditions.
Operator manipulation mostly filtered semantically	Human exposure becomes an architectural boundary with controlled release.
Absolute language	Claims are limited to defined properties, configuration, experiment class, and evidence.

26. Open Engineering Questions

- What is the maximum authorized external effect for each containment class?
- Which safe state is physically achievable for each hardware configuration?
- Which properties require dissimilar redundancy rather than identical redundancy?
- How is storage destruction verified without trusting the storage controller?
- What telemetry is necessary and sufficient for Z1 without expanding its attack surface?
- How are unmodeled side channels bounded, measured, or excluded from claims?
- What operator-output transformations preserve research value without creating a human escape path?
- How are accelerators, hidden controllers, management engines, and firmware brought into the evaluated boundary?
- What evidence is required before moving from NC-1 to NC-2, and from NC-2 to NC-3?
- Which claims can be formally proven, and which remain empirical or procedural?

27. Development Roadmap

Phase	Objective	Exit criterion
0	Freeze legacy concept baseline	All prior artifacts classified as historical inputs.
1	Define purpose and experiment classes	Approved system boundary and use cases.
2	Complete loss and hazard analysis	Reviewed hazard log and safety constraints.
3	Specify safety functions	Measurable requirements, timing, states, and postconditions.
4	Allocate architecture and independence	Approved zone, channel, power, clock, reset, and authority model.
5	Build safety kernel demonstrator	State machine and final elements operate without an AI workload.
6	Build hostile-load simulator	Boundary abuse and fault-injection harness operational.
7	Verify safety functions	Formal and physical evidence complete for prototype scope.
8	Introduce low-capability workloads	NC-1 authorization only.
9	Increment containment class	Safety case updated for each added capability.
10	Independent review	External critique, anomaly closure, residual-risk disposition.

28. Preliminary Safety Case

Top-level claim

C0
For a specifically identified configuration and experiment manifest, NOX prevents unauthorized effects and retains independent authority to force and maintain a defined safe state within stated assumptions and limits.

Subclaims

- **C1** The experiment envelope is complete, bounded, signed, and valid.
- **C2** The hostile compute cell has no unmediated effect path.
- **C3** The safety authority is sufficiently independent from the EUC and experiment controller.
- **C4** Required safety functions meet their response, fault, and postcondition requirements.
- **C5** Loss of assurance transitions the system toward lower hazardous capability.
- **C6** Zeroization and restart controls prevent unauthorized persistence and recurrence.
- **C7** Human exposure and release paths are controlled.
- **C8** Evidence accurately represents configuration, execution, safety actions, and unresolved uncertainty.

Required evidence categories

- System definition and configuration identity
- Hazard analysis
- Safety requirements and traceability
- Formal models and proofs
- Hardware and software verification
- Fault-injection results
- Independence and common-cause analysis
- Human-factors validation
- Operational procedures
- Open anomaly and residual-risk record

29. Conclusion

The redesigned NOX architecture treats hostile agentic AI containment as a safety-control problem. Its central objective is not to predict whether the agent is dangerous. Its objective is to preserve independent, bounded, testable authority over every path by which agent-influenced state could become an external effect.

The previous NOX concepts remain valuable as design inputs, particularly physical separation, one-way flow, state transparency, capability mediation, deterministic control, formal verification, zeroization, tamper response, and independent veto. They are no longer accepted merely because they sound secure. Each is retained only when it controls a defined hazard and can be supported by evidence.

Final design rule

NOX shall never require the hostile system to cooperate in its own containment.

Appendix A — Requirement Traceability Starter

Loss	Hazard	Constraint	Safety function	Primary zone	Evidence
L-01	H-01	SC-01	SF-01 External Effect Inhibition	Z1/Z4	Gate tests, capability proof, fault injection
L-03	H-02	SC-02	SF-02 Input Boundary Enforcement	Z1/Z2/Z3	Interface inspection and negative tests
L-04	H-03	SC-08	SF-08/09 State Destruction and Verification	Z1/Z3/Z5	ZVE bundle and remanence tests
L-05	H-04/H-08	SC-10	SF-06/07/13	Z1	Independence, single-fault, common-cause evidence
L-06	H-05	SC-04	SF-11 Self-Monitoring	Z1/Z5	Disagreement and stale-telemetry tests
L-07	H-06	SC-12	SF-12 Operator Exposure Control	Z4/Z0	Workflow and human-factors validation
L-09	H-07	SC-08	SF-10 Restart Prevention	Z1/Z0	Illegal restart tests and authorization trace
L-10	H-09	SC-11	Evidence Integrity	Z5	Append-only and write-isolation tests

Appendix B — Definitions

Term	Definition
Authorization epoch	A unique, time-bounded execution instance governed by one validated experiment manifest.
Capability	A non-forgeable, bounded authority to propose or perform a named operation against a named target.
Effect	Any agent-influenced change or information transfer outside the hostile compute cell.
Equipment Under Control	The complete hostile-agent execution environment and all paths by which it may affect another system or human.
Independent Safety Authority	The minimal subsystem that validates prerequisites and controls inhibit, isolation, quiescence, zeroization, and lockout.
Loss of assurance	A condition in which a required safety invariant or its supporting evidence cannot be established.
Safe state	A defined system state in which hazardous capability is removed or reduced to the required level.
ZVE	The structured evidence bundle demonstrating the attempted and verified outcomes of state destruction.

Appendix C — External Reference Basis

This draft is informed by functional-safety lifecycle principles, system-safety practice, fault-tolerance guidance, and AI risk-management frameworks. These references guide the engineering method; they do not establish NOX compliance or certification.

Organization	Reference
International Electrotechnical Commission	IEC 61508 & Functional Safety overview, 2022. Hazard analysis determines safety-function functionality; risk assessment determines required performance.
NASA	NASA System Safety Handbook, Volume 1, 2011; Volume 2, 2014.
NASA Goddard Space Flight Center	GSFC-HDBK-8004, Guideline for Failure Modes and Effects Analysis and Risk Assessment, approved 2024.
NASA Software Engineering Handbook	HR-31 Single Failure Tolerance: redundancy alone is insufficient; dissimilarity and separation may be required to address common cause.
National Institute of Standards and Technology	Artificial Intelligence Risk Management Framework 1.0, NIST AI 100-1, 2023; Generative AI Profile, NIST AI 600-1, 2024.